

Focus Shield AI

Conquering Multilingual Chaos: Autonomous AI Triage for Global Finance

UiPath AgentHack. Build the AI agents of tomorrow.



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Problem statement and proposed solution

Problem

What real-world problem are you solving?

- Overwhelming Multilingual Invoice Volume:**

Processing massive monthly invoice PDFs across diverse formats and languages (JP/EN/DE/ES/TH).

- Inefficient Manual Labor & Cognitive Fatigue:**

Manually searching, transcribing, and validating data entries drains the staff's focus, leading to severe mental fatigue and a higher risk of human error.

- Flawed Double-Checking & Dwindling Resources:**

Double-checks based on unverified comparison. Consequently, both preparers and reviewers incur immense verification costs, leading to a vicious cycle of compromised quality and diminished motivation."

Why does it matter?

- Disruption of Core Responsibilities:**

Non-core tasks squeezed into daily gaps act as major operational 'noise'—stripping back-office personnel of time for creative and strategic work.

- The Limit of Scale and Speed:**

Manual & conventional review alone cannot handle high-volume, multi-language, and high-frequency demands, creating a critical bottleneck for both organizational scale-up and data reliability.

- The Need for an "AI-Led Shield":**

Unsustainable manual full-volume verification should shift to AI-driven triage, building a robust shield so that people focus on deciding rather than checking.

Solution

Brief summary of the solution and what makes it unique

- Focus Shield AI: An Autonomous AI Triage System** that

cross-examines multi-AI data, isolating only mismatches and anomalies for human judgment. By automating error-free invoices, it liberates teams from full-volume reviews to focus entirely on critical decision-making.

High-level value prop or innovation angle

- Revolutionary "Multi-AI Consensus":**

Breaks the traditional mold of "AI extracts, human checks everything." By cross-referencing insights from distinct AI models, the system autonomously filters out noise and misrecognitions.

- Seamless Multilingual Adaptation:**

A resilient global finance foundation that neutralizes the complexities of diverse languages, currencies, and formats.

- Drastic Reduction of Mental Burden:**

Personnel interact strictly with exceptions. Eliminating operational uncertainty directly ensures team morale and psychological safety.

- Uncompromised Scalability:**

Scales invoice volumes indefinitely without consuming human resources, simultaneously maximizing processing speed and data accuracy.

Benefits and technologies used

Details

End-user	Accounting Personnel
User department	Accounting
Industries	Manufacturing
UiPath products used	Studio, IXP , Orchestrator, Action Center, UiPath Agents, Maestro, Data Fabric, Integration Service
Other - integrations / APIs / technologies used	LLM extraction API, OCR engines, SQL database integration, OneDrive, REST API integration

Benefits, impact and outcomes

What does this agent actually achieve?

Autonomous Triage to Eliminate Full Reviews:

Automatically cross-references data across multiple AIs, seamlessly escalating only true anomalies and mismatches to human stakeholders.

True Straight-Through Processing (STP):

Flawless data bypasses human intervention entirely, registering instantly into Data Fabric and eradicating bulk verification workflows.

Cognitive Load Minimization & Absolute Focus:

Wipes out operational noise so that team members can allocate 100% of their bandwidth to strategic exceptions requiring human genius.

Elevated Employee Engagement & Well-being:

Removes the mental burnout of manual double-checking and the anxiety of human error, fostering a sustainable, psychology-safe work culture.

Time saved, steps automated, user pain reduced, etc.

Dramatic Time Savings (Up to 90% reduction):

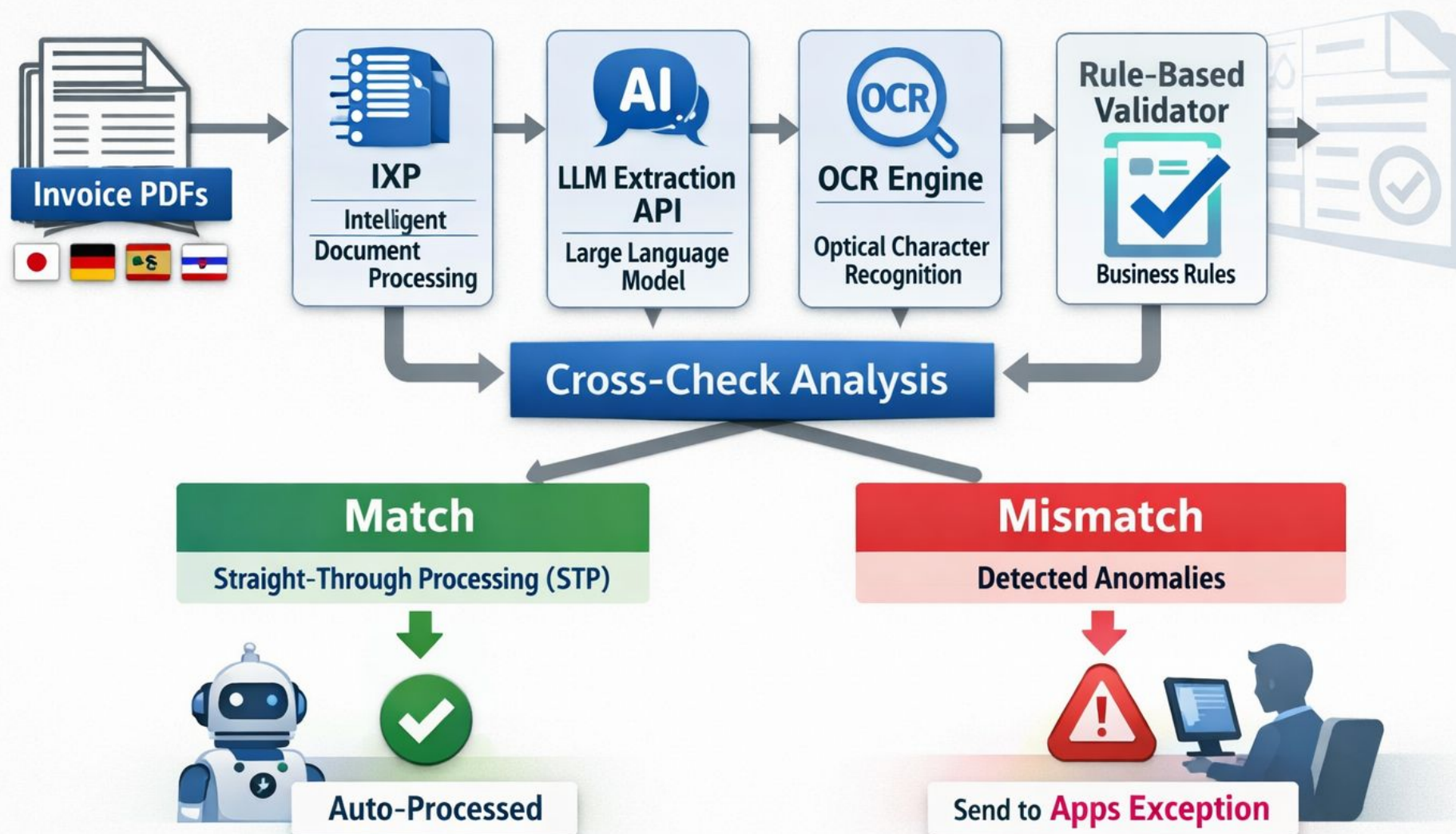
Autonomous processing shrinks the monthly 2–3 days spent on manual invoice checks to just a few hours, entirely reinvesting the saved time into creative, strategic work to design the organization's future.

Shifting from "Defensive Efficiency" to "Offensive Innovation" (100% Scalability):

Even if invoice volumes or languages double or triple, human workload remains flat—wiping out routine noise to free up mindshare and directly scale corporate innovation.

Multi-AI Consensus Architecture

Cross-Checking Intelligent Extraction



Manual Verification → Autonomous Triage (Before / After)

BEFORE

Manual Invoice Checks



Manual Verification
→
Autonomous Triage



2-3 Days

AFTER

Automated Processing



Few Hours

Full Human Review



AI Processing



99% Auto-Processed

Human Exceptions

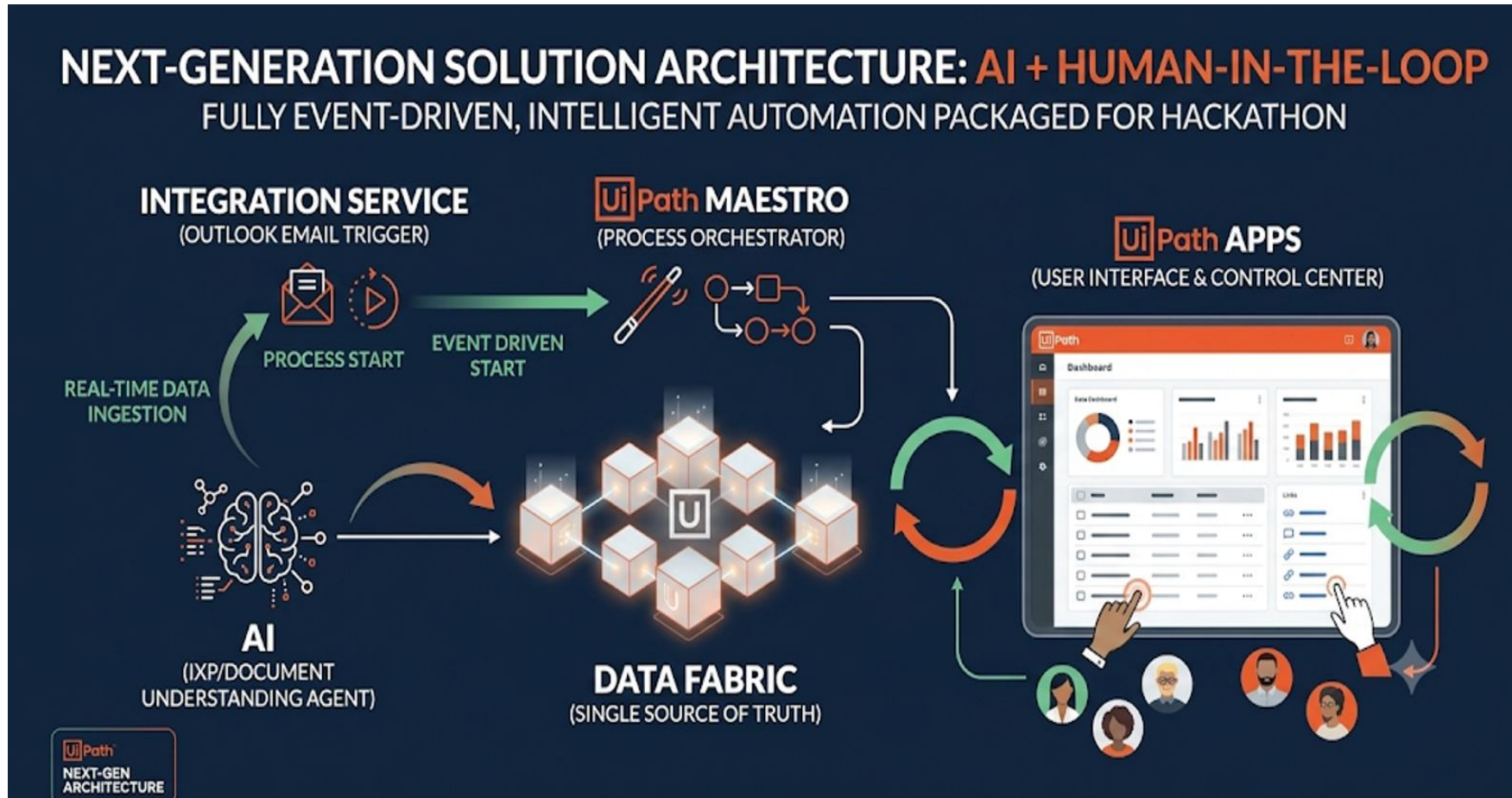


1% Manual Review

Solution architecture– Agentic Automation Overview



Solution Architecture Overview



This solution features a fully event-driven, intelligent automation architecture that combines **UiPath's next-generation components** (Integration Service, Maestro, Data Fabric, and Apps) with **AI (IXP/Document Understanding Agent)**.

By utilizing Data Fabric and Apps as the central data and user layer, it achieves seamless **Human-in-the-Loop** collaboration, allowing humans and AI to co-process data in real-time.

Solution architecture– Agentic Automation Overview

Technology Stack & Roles

Layer / Tool	Role & Hackathon Positioning
Outlook / Integration Service	【Trigger】 Detects incoming emails and instantly triggers the process.
UiPath Orchestrator Queue	【Dispatcher & Buffer】 Manages queues for concurrent processing and error handling, built for high scalability.
UiPath Maestro	【Performer】 Acts as the main engine, triggered by the queue to orchestrate the entire end-to-end workflow.
IXP (DU Agent)	【AI Brain】 An intelligent agent capable of high-accuracy invoice data extraction and language detection.
UiPath Data Fabric	【Single Source of Truth】 Unified data layer for real-time tracking of processing statuses and AI extraction results.
UiPath Apps	【Control Center】 A unified UI providing a bird's-eye view of the system, allowing users to jump to specific records via links to verify data.

Solution architecture– Agentic Automation Overview

End-to-End Data & Process Flow

These are the detailed steps of how the system processes data.

Phase / Component	Technical Execution Details
1. Ingestion & Queueing (Dispatcher)	• The Integration Service (Email Trigger) detects an incoming invoice (PDF) in Outlook. • It extracts the PDF file and registers it as a single queue item (one record per file) into the Orchestrator Queue (Dispatcher).
2. Process Initiation (Maestro Start)	• The Queue Trigger automatically fires up the Performer bot built using UiPath Maestro. • [Add Record] As a starting event, an initial record is immediately created in the Data Fabric entity to track the process.
3. AI-Powered Invoice Parsing (IXP)	• [Read Invoice] The file is passed to the IXP (Document Understanding Agent). • The AI extracts the invoice data and performs a Language Check (Japanese / English / Other).
4. Validation & Human-in-the-Loop	• [Decision] The system runs validation checks based on confidence scores and required fields. ✗ If validation fails (Defect Found): A validation task is sent to UiPath Apps. The user views the dashboard, clicks the detailed link to inspect the specific data, and makes corrections. Once fixed, the Record is Updated (Action URL triggered), routing it back to the main path. ○ If validation passes (No Defect): It bypasses human intervention and proceeds directly to the language routing logic.
5. Execution & Completion	• [Language Routing] The business logic branches dynamically based on the language classified by the AI (Japanese, English, or Other). • [Store File] The processed file is securely saved to the designated storage. • [Update Record] The final transaction data and a "Completed" status are written to the Data Fabric entity.

**THANK
you.**

Silence the Invoice Noise. Empower Human Genius